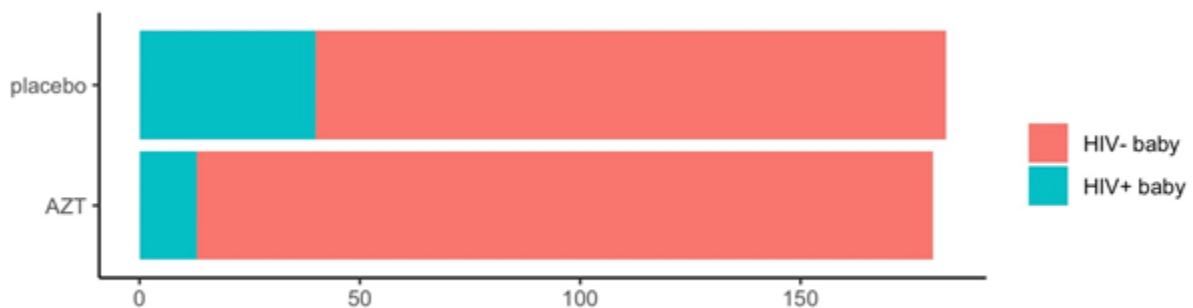


MATH 213 -- Quiz 6

Names of group members: Moyed , Daniel

In 1993, in one of the first studies about preventing transmission of HIV infection from mothers to babies before and during birth, researchers gave the drug AZT to pregnant, HIV-infected women (Connor et al., 1994). Roughly half of the women in the experiment were randomly assigned to receive the drug AZT, and the others received a placebo. The HIV-infection status was then determined for the babies. The results of the experiment are summarized in the table and graph below.

	Placebo treatment	AZT treatment	Total
HIV-infected baby (HIV+)	40	13	53
Non-HIV-infected baby (HIV-)	143	167	310
Total	183	180	363



1. What are the explanatory and response variables for this experiment? Classify the variables as categorical or numerical.

Explanatory variable: whether drug(AZT) was received

Type (categorical or numerical):Categorical

Response variable: if the baby has HIV or not

Type (categorical or numerical): Categorical

2. Why did the researchers assign the treatment (AZT or placebo) at random? Explain using terminology from earlier this semester.

The researchers assigned the treatment (AZT or placebo) at random so they can generalize results and determine causation. If they randomly choose, everyone has an even chance of receiving either treatment which eliminates sampling bias. And randomly assigning the treatments can determine if there were placebo effects in the subjects.

3. Describe what you learn from the data, mentioning the difference in proportions of HIV+ babies in the two groups. (Calculate that number!) What preliminary conclusions do you draw from this experiment after looking at the summary of the results in the graph and table?

Preliminary conclusions: The data shows that 93% of people in the treatment group who received the drug (AZT) showed signs of their children not having HIV. and those in the group that received a placebo had a 78% of not having HIV. The group who received the treatment had a 15% higher chance of their baby not having HIV.

You will investigate next the hypothesis that the difference between the AZT and placebo groups is simply a result of the randomization into the assigned groups.

4. Write your null and alternative hypotheses, both in words and using the correct notation.

Null hypothesis(H_0): Receiving the treatment(AZT) is no better than receiving the control(placebo)

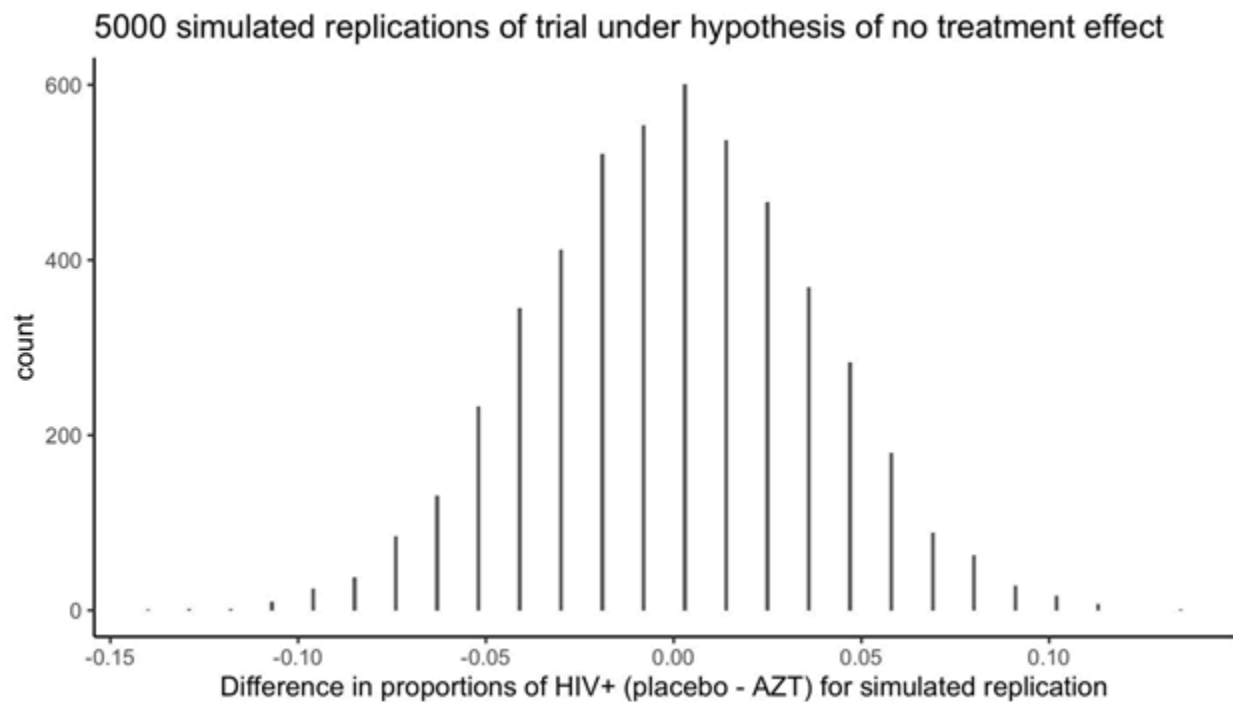
Alternative hypothesis(H_1): Receiving the treatment(AZT) will have a higher cure rate than receiving a placebo.

Notation:

$H_0: p = 0$ (no difference in HIV+ rates)

$H_1: p < 0$ (AZT has a lower infection rate)

I used R to simulate repeating the experiment 5000 times. The following histogram summarizes these 5000 simulated replications.



5. What does each dot on this histogram represent? Be specific about the assumptions that underlie these simulations.

Each dot represents a simulation of 383 women assuming that the treatment had no effect on cure.

6. What do you conclude from my 5000 simulated replications? Explain your conclusion carefully, referring to appropriate details of the histogram.
The simulation shows that very few (or none) of the 5000 randomized trials produced a

difference in proportions as large as the one observed in the actual data. This suggests that the observed difference is unlikely to be due to chance alone. Therefore, we have strong evidence to reject the null hypothesis and conclude that AZT treatment likely reduces the chance of HIV transmission.